



UNIVERSITÀ DEGLI STUDI DI TRENTO

Dipartimento di Ingegneria
e Scienza dell'Informazione

Fundamentals of Robotics

Introduction

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What are robots?

- Robot is a Czech word, *robota*, meaning "forced labour"
- The word was introduced by a novelist, Karel Čapek, in 1920 and became famous for the vast body of Science Fiction that it generated
- More recently the word refers to a well defined technological domain, which over the course of the last years, has generated a multi-billion euros worth market

A definition

- The Longman dictionary entry for robot is

A machine that can **move** and do some of the work of a person
and is usually **controlled by a computer**

- The stress is on the ability to move and on the system intelligence



Example 1

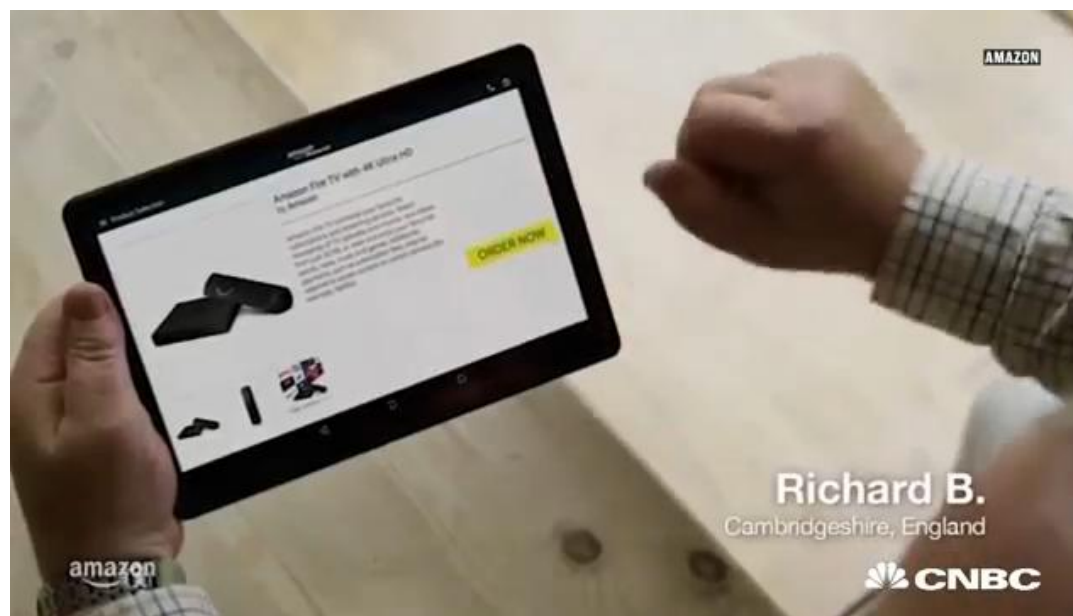
- Robotic Arm
 - See <https://www.youtube.com/watch?v=y3e1P-OMq0Q>
 - It moves to operate on the manufactured piece
 - Its motion is controlled at different levels
 - Local: motion controllers on the joints
 - Global: motion plan to execute





Example 2

- Drone
 - see <https://www.youtube.com/watch?v=MR9PoBAssw0>
 - It can fly (autonomously or controlled) for surveillance, delivery or just for fun
- Control levels
 - Motors (rpm of each motor)
 - Trajectory following (on board)
 - Mission planning (remote)





Example 3

Food Delivery robot

- See <https://www.youtube.com/watch?v=cZTCmx6N7Xc>
- Control levels
 - Motors
 - Path
 - Mission





Example 4

- A quick look to “made in Trento” assistive technology

<https://www.youtube.com/watch?v=haYAwUh68ml>

The key is understanding and
adapting to human needs



Example 5

What about making the whole things a little crispier by inserting some competition?

- Control all levels + try to adapt to the adversary intent in order to win the race

[Autonomous racing - YouTube](#)

Nice combination of game theory, graph theory and optimal control



Example 6

- Human as pathfinder
 - The robot follows closely a human elected as «guide»

[Human as pathfinders for safe navigations - YouTube](#)

Big issue: identifying and tracking your pathfinder!!!



A few observations

- The robots shown in the previous examples have the following abilities
 - They estimate their position (more generally their state) in the environment
 - They can sense the environment and create a map
 - They take a decision
 - They act to implement the decision



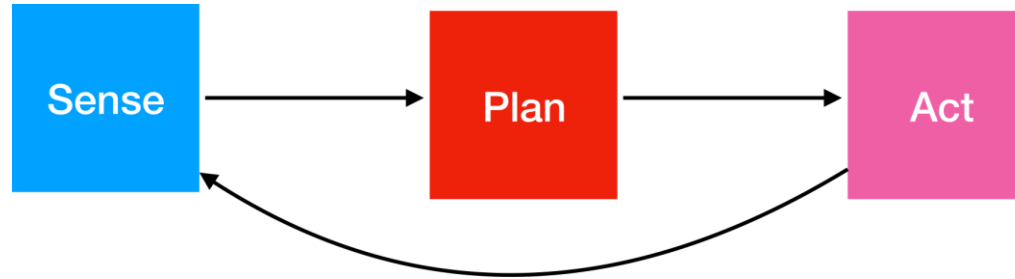
The system intelligence

- The system intelligence is concentrated in two different aspects
 - **Perception** (surrounding environment, and ...itself in the environment)
 - **Decision** (what to do and how to do it)
- Our course will revolve around these two aspects

Architectures



Deliberative Paradigm



- Top/Down philosophy
- The Robot
 1. Senses the world and constructs an updated model of the environment
 2. Plans a course of action
 3. Implements the plan

Assumption: The environment is closed and its only changes are due to the robot action

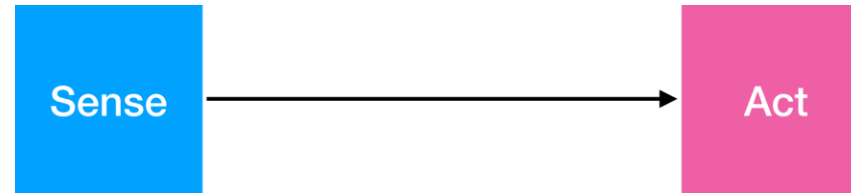


Deliberative Paradigm

- In this paradigm all decisions are taken after a thorough deliberation procedures
- Planning is prominent because we will not make corrections before the next sensing operation
- We need a comprehensive model of the environment
- The responses of the system are therefore quite slow



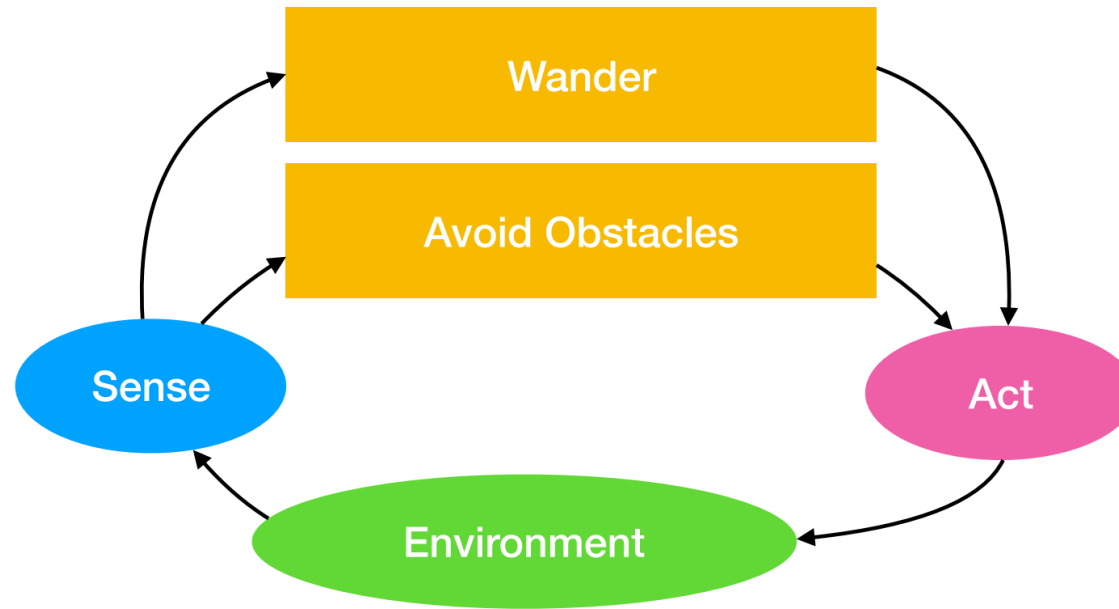
Reactive Paradigm



- Sens-Act organisation
- A reaction in response to a sensed data
- Several Sensing/Reaction pairs are stored (system behaviours), which operate concurrently
- The robot will actually execute a combination of behaviours



Reactive Paradigm



- **Situated** agent, robot is integral part of the world.
- **No memory**, controlled by what is happening in the world.
- **Tight coupling** between perception and action via behaviours.
- Only local, behaviour-specific sensing is permitted (**ego-centric** representation).
- The overall behaviour of the robot emerges from the combination of the different behaviours



Example Potential Fields

- Treat the robot as a **particle** acting under the influence of a potential fields
- The Robot travels along the derivative of the potential
- The field depends on obstacles, desired travel directions and targets
- The Resulting field (vector) is given by the **summation of primitive fields**
- The strength of the field may change with distance to obstacle/target

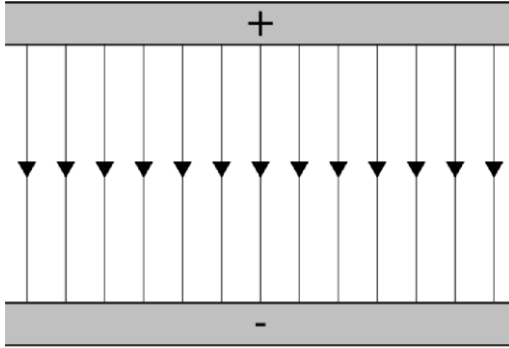


Example Corridor Following

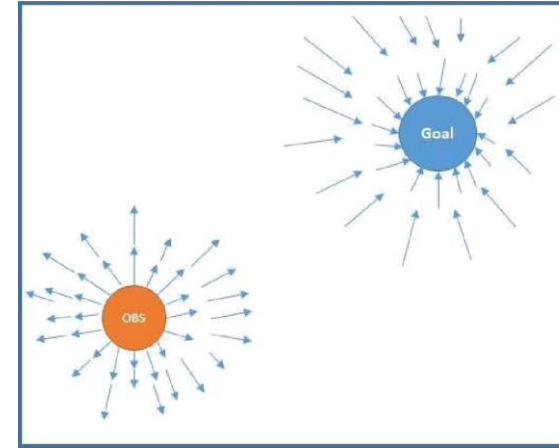
- **Level 0** (collision avoidance) is done by the repulsive fields of detected obstacles.
- **Level 1** (wander) adds a uniform field.
- **Level 2** (corridor following) replaces the wander field by three fields (two perpendicular, one uniform).



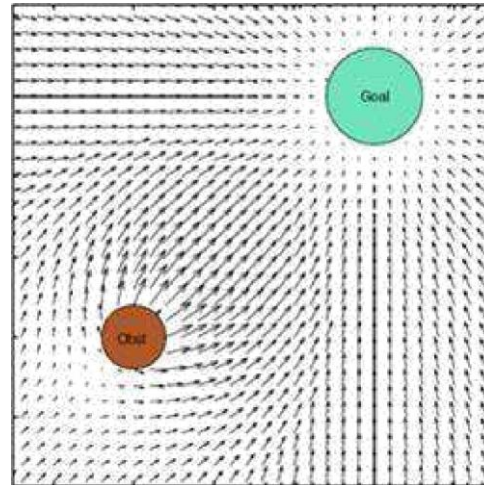
Potential Fields



Uniform Potential Field



Attractive/Repulsive Fields



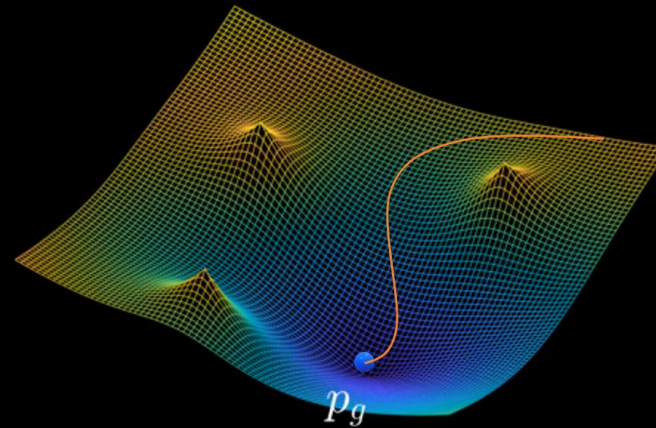
Combination

An Example

Navigation via Potential Fields for Nonholonomic vehicles

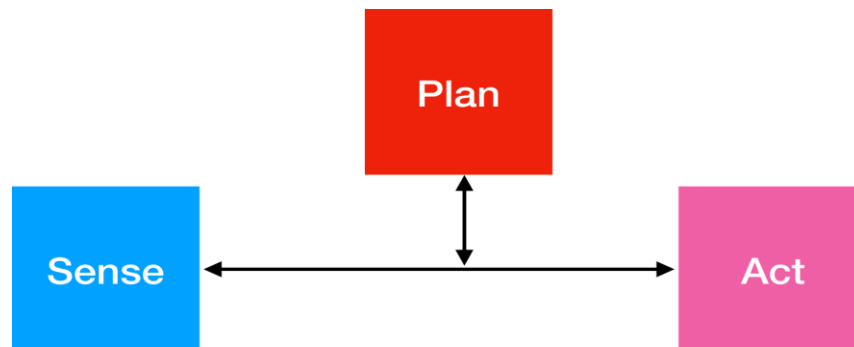
Since we assume to know the environment, we compute a safe path before the guidance starts.

By discretising it in way-points and by changing the potential function according to the active waypoint, we are able to avoid local minima generated by static obstacles in the map.





Hybrid Paradigm



- The robot first plans (deliberates) how to best decompose a task into subtasks (also called “mission planning”) and then what are the suitable behaviours to accomplish each subtask.
- Then the behaviours starts executing as per the Reactive Paradigm.
- Sensing organisation is also a mixture of Hierarchical and Reactive styles; sensor data gets routed to each behaviour that needs that sensor, but is also available to the planner for construction of a task-oriented global world model.



Hybrid Paradigm

- Hybrid Paradigms combine a good attention for planning with the ability for the system to react to unexpected changes in the environment
- The idea is to have combination of strategy (for planning) and tactics (for the execution of subtasks)
- The system maximises performance paying the price of a high level of complexity.



Perception

- In the context described above perception is key
- ...but what is perception?

Perception is a cognitive activity that makes the system aware of itself and of its environment

- It helps the robot respond such queries as

How can I detect and classify the different entities of my surrounding environment

How can I construct a map that contains both metric and semantic information?

Where am I in the map right now?

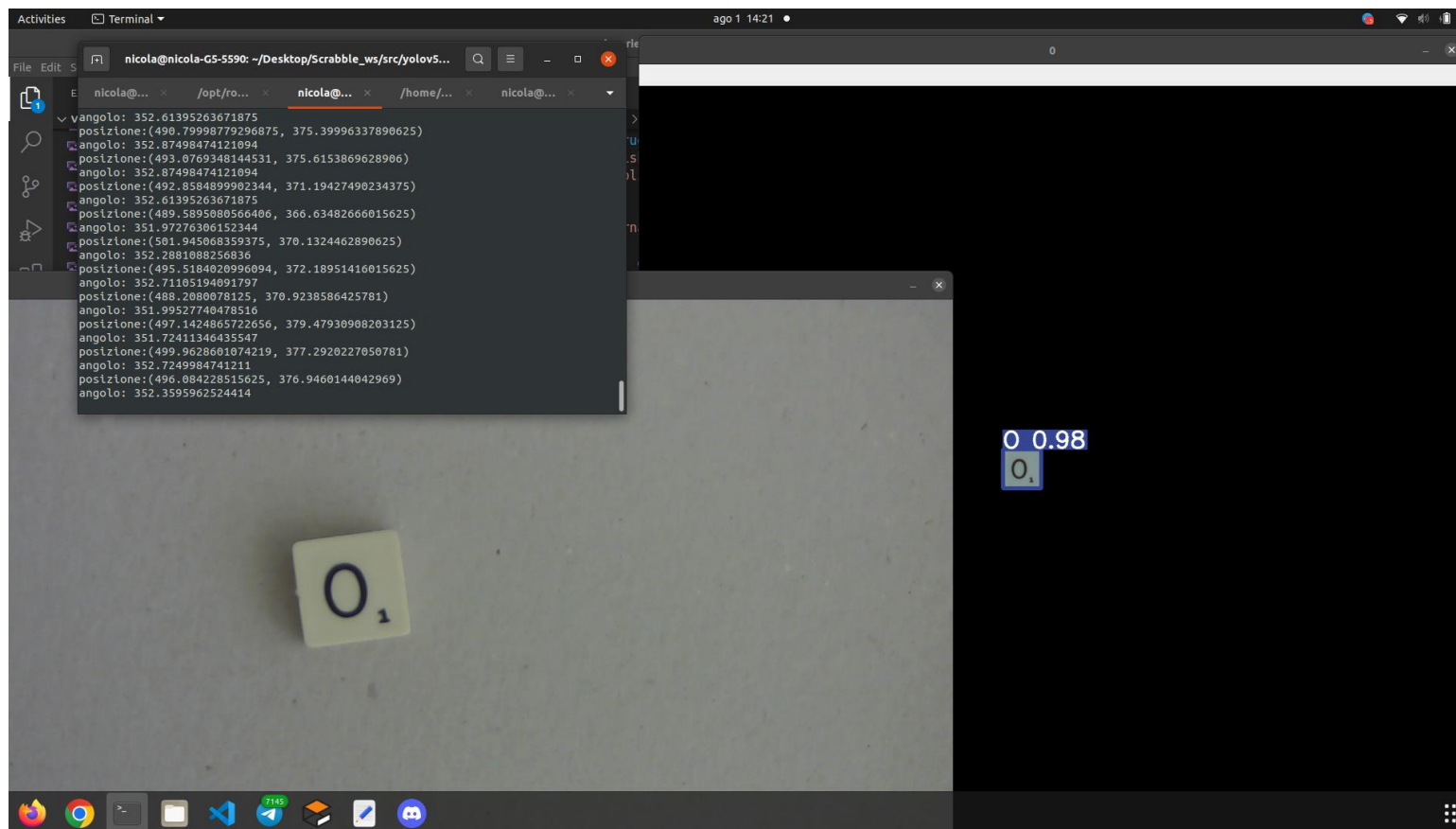
Where are the objects I need to interact with or avoid?

- This information is key to take the right decisions both in terms of which action I should do and of which path I should follow.



Examples

- Example from the scrabble game





Perceiving humans

- Humans are increasingly part of the robot's horizon
- Sometimes the robot is required to operate in an environment populated by humans and has to preserve their safe
- Sometimes the robot is supposed to collaborate with the human (e.g., cobots)
- Sometimes the robot is supposed to assist the human
-Once again a reliable perception is key

Where are the humans in the scene
and where are they going?
(SAFETY)

What is the human 's affective state
right now?
(ASSISTANCE)

What action is a human doing right now,
and what will she/he do in the near future?
(COLLABORATION)

- In the course we will provide an overview on the main techniques used for robot perception and will put our hands in the dirt with some of them.



Example

- Gesture recognition for HRI (provided by one of your former colleagues)





Course contents

....almost time to peek into the course contents...



Foreword

1. The course adopts a very practical approach
 - You will receive an introduction to the most important theoretical foundations
 - We will have exercises and practical sessions
 - You will develop experience with simulators and actual systems in the laboratory
2. You will learn to program complex robotic systems using state of the art frameworks (the ROS middleware)

Grading		
	1. Written test	
	2. Oral discussion of the test	
	3. Project Presentation	

Mandatory: enough to achieve 27/30

Optional: needed for grade in the range 27-30



Course Content

- Theory and Practice of Robotic Manipulation
 - ^e Modelling robots (2 hours)
 - Geometry Background (7 hours)
 - Forward kinematics (5 hours)
 - Inverse Kinematics (5 hours)
 - Differential Kinematics (7 hours)
- Modelling the UR5 manipulator
 - Modeling (3 hours)
 - Simulation experience (2 hours)
- Advanced Topics
 - Advance topics (2 hours)
 - Exercises on advanced topics (2 hours)



Course Contents

- Perception
 - Sensing devices (3 hours)
 - Robot Vision (3 hours)
 - Object Recognition (6 hours)
 - Biometrics/Face recognition (3 hours)
 - Humanoid Robots: ARI (3 hours)
 - SLAM (3 hours)
 - Localisation (3 hours)
 - Template Matching (2 hours)